

Final Project Report

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Introduction

Wildfires represent a growing threat to ecosystems, communities, and infrastructure across the United States. Understanding what drives and causes the total area burned in a given month, and whether early-season fire activity can serve as a leading indicator, and has practical value for resource allocation and risk management.

This analysis uses the US Wildfire dataset from the USDA in conjunction with the US Forest Service, which contains records of wildfires reported across the US from 1992 to 2020. Each record includes the fire discovery date, containment date, fire size (in acres) geographic location, and cause classification. We aggregate these records to monthly summaries at the national level, producing two functional observations per year: a monthly fire count curve and a monthly mean area burned curve.

We ask two scientific questions:

1. Does the monthly profile of fire counts across a year predict the monthly profile of mean area burned that same year?
2. Does restricting predictions to only use past fire counts improve or degrade predictive performance compared to using the full-year fire count profile?

To answer these questions we apply the **historical functional linear model** of Malfait and Ramsay (2003), which models a functional response as a smooth function of only the past history of a functional predictor. This is a natural fit for our setting because it does not make scientific sense to use fire counts from a later month to predict area burned in an earlier month. We compare this to an unrestricted function-on-function regression model as a baseline, both fitted using the `pffr` framework in the `refund` package for R.

Methods

Functional Data Setup

Let $i = 1, \dots, n$ index years and $t \in \{1, \dots, 12\}$ index months. For each year i we observe:

- $Y_i(t)$: mean area burned (acres) in month t
- $X_i(s)$: number of fires recorded in month s

We apply a $\log(x + 1)$ transformation to both $Y_i(t)$ and $X_i(s)$ to reduce skewness before modeling.

Historical Functional Linear Model

Malfait and Ramsay (2003) introduced the historical functional linear model for settings where a functional predictor $X_i(s)$ influences a functional response $Y_i(t)$, but only through its past values. The model is:

$$Y_i(t) = \alpha(t) + \int_{\mathcal{S}(t)} \beta(s, t) X_i(s) ds + \varepsilon_i(t)$$

where the integration domain $\mathcal{S}(t) = \{s : s < t\}$ enforces the historical constraint, $\alpha(t)$ is a smooth intercept function, $\beta(s, t)$ is a bivariate coefficient surface defined only on the region $s < t$, and $\varepsilon_i(t)$ is a mean-zero error process. The surface $\beta(s, t)$ captures the influence that fire counts at month s have on mean area burned at month t , restricted to cases where s precedes t .

We estimate this model using the `pffr` function from the `refund` package in R, which represents $\beta(s, t)$ using penalized bivariate splines and selects smoothing parameters via neighborhood cross-validation (NCV). The historical constraint is specified via `limits = 's<t'` in the `ff()` term.

Comparison Model

As a baseline we also fit the unrestricted function-on-function regression model:

$$Y_i(t) = \alpha(t) + \int_{\mathcal{S}} \beta(s, t) X_i(s) ds + \varepsilon_i(t)$$

where integration is over the full domain $\mathcal{S} = \{1, \dots, 12\}$. This model is scientifically less appropriate since it allows future fire counts to inform predictions of past area burned, but serves as a useful upper bound on in-sample fit.

Prediction and Evaluation

We split the data into a training set (1992–2015, $n = 24$ years) and a test set (2016–2020, $n = 5$ years). The historical model test predictions are computed manually using the estimated intercept $\hat{\mu}(t)$ and coefficient surface $\hat{\beta}(s, t)$:

$$\hat{Y}_i(t) = \hat{\alpha}(t) + \sum_{s < t} \hat{\beta}(s, t) X_i(s)$$

Both models are evaluated using root mean squared error (RMSE) on the held-out test years, computed on the original (non-log) scale. We also computed the RMSE on the original.

Data Analysis

Unrestricted Model

```
FullModel <- pffr(  
  Y_log ~ ff(X_log, xind = s),  
  yind = t,  
  method = "NCV",  
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))  
)
```

```
R2_reg <- summary(FullModel)  
# R-squared of .568
```

Historical Model

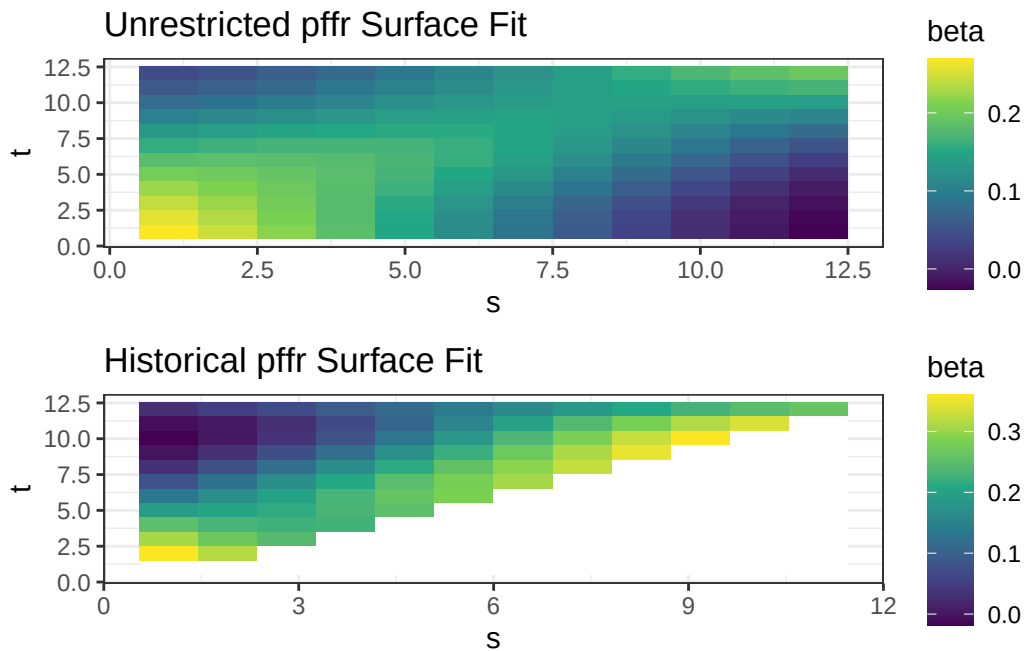
```
FullModel_Hist <- pffr(  
  Y_log ~ ff(X_log, xind = s, limits = 's<t'),  
  yind = t,  
  method = "NCV",  
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))  
)
```

```
R2_hist <- summary(FullModel_Hist)  
# R-squared of .6
```

Plotting the Models

using `seWithMean` for `s(t.vec)` .

using `seWithMean` for `s(t.vec)` .



Predictions

Unrestricted Train Model

```
Train_Model <- pffr(  
  Y_log_train ~ ff(X_log_train, xind = s),  
  yind = t,  
  method = "NCV",  
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))  
)
```

```
R2_full_train <- summary(Train_Model)  
# R^2 = 0.613
```

Historical Train Model

```
Train_Model_Hist <- pffr(  
  Y_log_train ~ ff(X_log_train, xind = s, limits = 's<t'),  
  yind = t,  
  method = "NCV",  
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))  
)
```

```
R2_hist_train <- summary(Train_Model_Hist)  
# R-squared of .637
```

Unrestricted Model Prediction

```
X_log_test <- log(X_fires_test + 1)  
reg.train_preds <- predict(  
  Train_Model,  
  newdata = list(X_log_train = X_log_test)  
)  
# Look at RMSE  
y_test <- t(Y_area_test)  
reg.train_preds_t <- t(reg.train_preds)  
regular_rmse <- sqrt(mean((y_test - reg.train_preds_t)^2))  
regular_rmse
```

```
[1] 141.591
```

Historical Model Prediction

```
# Get intercept  
smterms <- coef(Train_Model_Hist, n1 = 12, n2 = 12)$smterms[[1]]$value
```

using seWithMean for s(t.vec) .

```
pterm <- coef(Train_Model_Hist, n1 = 12, n2 = 12)$pterm[1, 1]
```

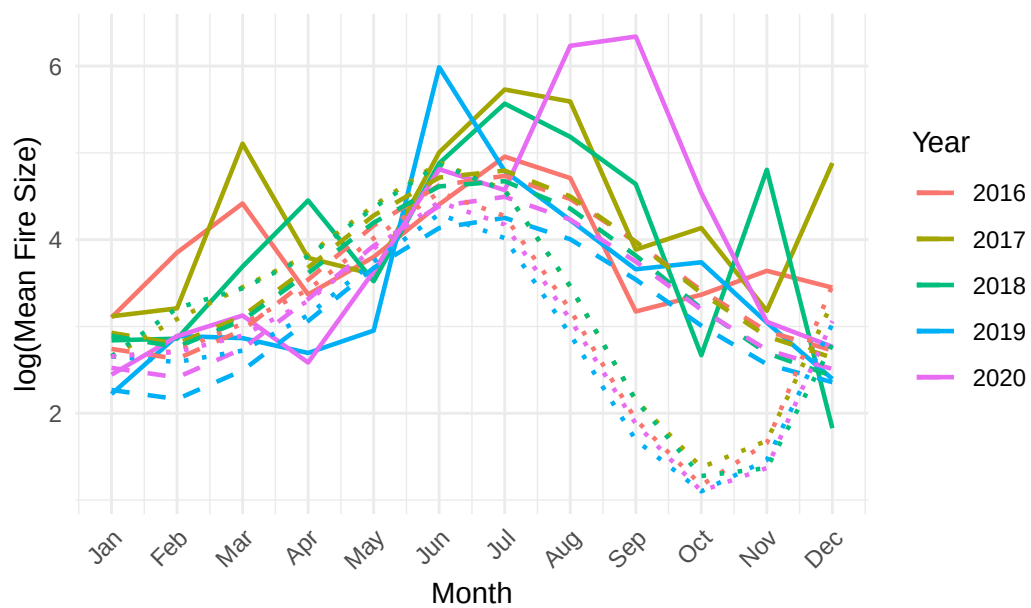
```
using seWithMean for s(t.vec) .
```

```
b0 <- pterms + smterms
b0_matrix <- matrix(b0, nrow = 12, ncol = 5)
# Get the beta surface matrix (12 x 12)
b1 <- Train_HistBetas
b1 <- matrix(b1, nrow = 12, ncol = 12)
b1[lower.tri(b1, diag = TRUE)] <- 0 # taking out values where s>=t
# Manual prediction
b1_matrix <- X_log_test %*% b1
b1_matrix_t <- t(b1_matrix)
hist.train_preds <- b0_matrix + b1_matrix_t
## Get RMSE's
historical_rmse <- sqrt(mean((y_test - hist.train_preds)^2))
historical_rmse
```

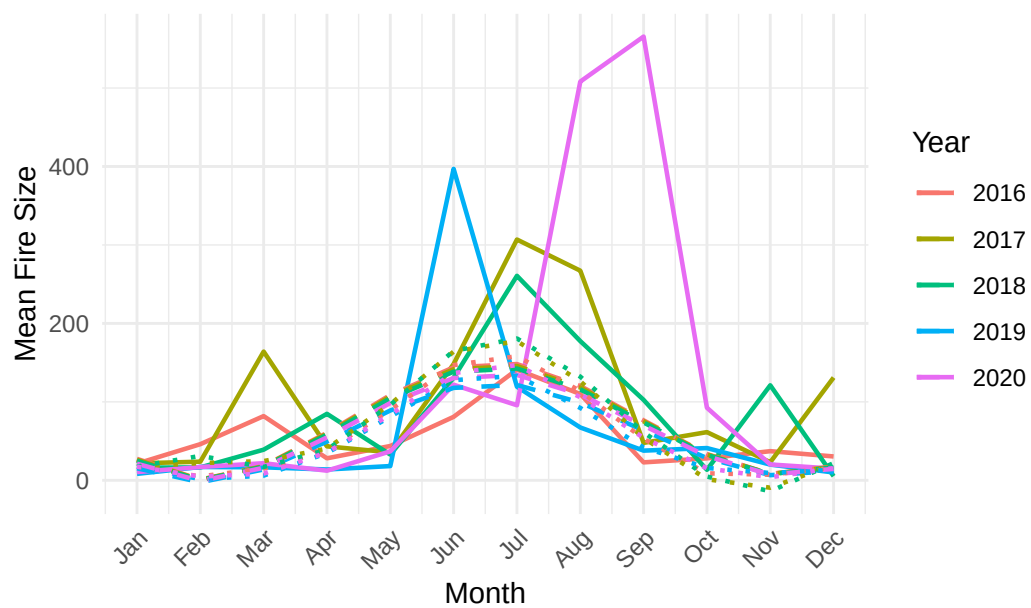
```
[1] 141.9687
```

Plot Predictions

Actual (solid) vs Unrestricted (dashed) vs Historical (dotted) Pre



Actual (solid) vs Unrestricted (dashed) vs Historical (dotted)



Conclusion

What we found:

Our analysis suggests that yes, monthly fire counts do predict mean area burned, with both models explaining a meaningful portion of the variability. The Unrestricted model achieved an R^2 of 0.568 on the full data and 0.613 on the training data, while the historical model performed slightly better at 0.60 and 0.637 respectively. This is noteworthy because the historical model uses strictly less information (only past fire counts), yet predicts better. This suggests the historical constraint actually helps the model generalize rather than hurting it.

The beta surface plots tell the most interesting story. In the Unrestricted function linear model plot, early season fire counts (January–March) are the strongest positive predictors of area burned, showing up as bright yellow in the bottom-left corner of both surfaces. This effect decays as the year progresses. The historical functional linear model plot shows that the same-month and near-future relationships are stronger predictors. However this also tapers off as the months go on.

For predictions on the held-out 2016–2020 test years, both models achieved similar RMSEs (regular: 141.6, historical: 141.9), meaning neither substantially outperformed the other in actual forecasting despite the historical model fitting the training data better.

When we did the log of the mean fire size we found that it really dipped in the later months of the year compared to when we did not take the log. This is likely due to a few factors. The main reason being that there is a real seasonal decline. Another one is that the log transformation just exaggerates the drop in general so if the drop went from 1000 to 200 it's going to be a much sharper drop on the log scale. Another factor that could be contributing is that at the end of the year there are a lot less fires so it becomes more noisy and more sensitive to different fire sizes. Finally the other thing is because of the spline boundary and it's smoothing effects, this smooths across months and at the edges like in December there is less data on one side. For predictions on the held-out 2016–2020 test years, both models achieved similar RMSEs (Unrestricted: 141.6, historical: 141.9), meaning neither substantially outperformed the other in actual forecasting despite the historical model fitting the training data better.

How certain we are:

We are moderately uncertain. We only have 29 years of data total, with just 5 test years, so the RMSE comparison is based on a very small test set and could easily flip with different train/test splits. The low effective rank warnings also suggest the fire count curves don't vary much in shape across years, which limits how much the functional predictor can really contribute beyond a scalar summary.

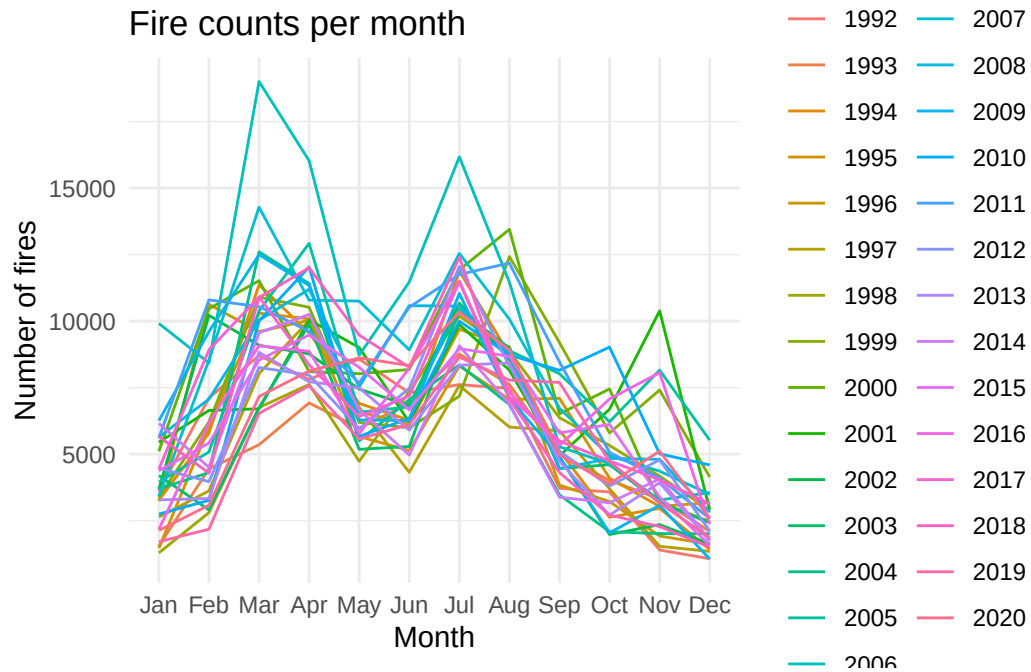
Follow-up questions worth exploring:

The most natural next question is whether adding climate variables like drought index, temperature, or precipitation as scalar predictors alongside the functional fire count predictor would improve predictions. Our plots show the model struggles most with extreme years like 2018 and 2020, which were anomalous fire seasons likely driven by climate conditions not captured in fire counts alone. We'd also ask whether predicting at a regional level rather than nationally would produce stronger relationships, since the national aggregation likely smooths out a lot of meaningful local variation.

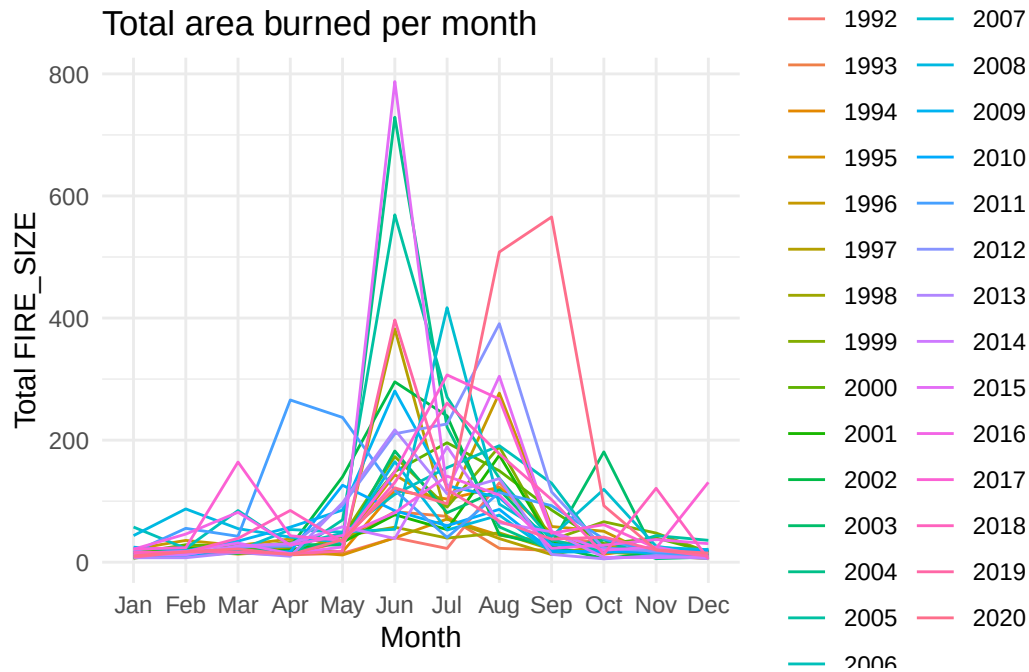
Appendix

Exploratory Analysis

```
ggplot(  
  fires_counts,  
  aes(x = month, y = n_fires, color = factor(year), group = year)  
) +  
  geom_line() +  
  labs(  
    title = "Fire counts per month",  
    x = "Month",  
    y = "Number of fires",  
    color = "Year"  
  ) +  
  theme_minimal()
```



```
ggplot(
  fires_month,
  aes(x = month, y = total_size, color = factor(year), group = year)
) +
  geom_line() +
  labs(
    title = "Total area burned per month",
    x = "Month",
    y = "Total FIRE_SIZE",
    color = "Year"
  ) +
  theme_minimal()
```



All Code

```
library(ggplot2)
library(tidyverse)
library(refund)
library(gridExtra)
library(patchwork)

fires <- vroom::vroom("forestfires.csv")
fires_counts <- fires |>
  mutate(
    year = FIRE_YEAR,
    month = month(DISCOVERY_DATE, label = TRUE)
  ) |>
  count(year, month, name = "n_fires")

fires_month <- fires |>
  mutate(
    year = FIRE_YEAR,
    month = month(DISCOVERY_DATE, label = TRUE)
  ) |>
```

```

group_by(year, month) |>
  summarise(total_size = mean(FIRE_SIZE, na.rm = TRUE), .groups = "drop")
# Making the dataset for our Y matrix
# Taking the values from month column and makes them each their own column
Y_dat <- fires_month |>
  pivot_wider(
    names_from = month,
    values_from = total_size
  )
# Then I want to make this a matrix without the year variable
Y_mat <- as.matrix(Y_dat)[, -1]
# Then do the same for the X matrix
# Create the dataset
X_dat <- fires_counts |>
  pivot_wider(
    names_from = month,
    values_from = n_fires
  )
# Then I want to make this a matrix
X_mat <- as.matrix(X_dat)[, -1]
t <- 1:ncol(Y_mat) # response index (months)
s <- 1:ncol(X_mat) # predictor index (months)
# For total area burned take the log
Y_log <- log(Y_mat + 1)
# For fire counts take the log
X_log <- log(X_mat + 1)
FullModel <- pffr(
  Y_log ~ ff(X_log, xind = s),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)
FullBetas <- coef(FullModel, n1 = 12, n2 = 12)$smterms[[2]]$value
R2_reg <- summary(FullModel)
# R-squared of .568
FullModel_Hist <- pffr(
  Y_log ~ ff(X_log, xind = s, limits = 's<t'),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)
FullHistBetas <- coef(FullModel_Hist, n1 = 12, n2 = 12)$smterms[[2]]$value

```

```

R2_hist <- summary(FullModel_Hist)
# R-squared of .6
FullModel2 <- coef(FullModel, n1 = 12, n2 = 12)$smterms[[2]]$coef
FullModel2_Hist <- coef(FullModel_Hist, n1 = 12, n2 = 12)$smterms[[2]]$coef
# Creating the dataframe
FullBetasdf <- data.frame(
  't' = c(FullModel2$X_log.tmat, FullModel2_Hist$X_log.tmat),
  's' = c(FullModel2$X_log.smat, FullModel2_Hist$X_log.smat),
  'beta' = c(FullModel2$value, FullModel2_Hist$value),
  'Fit' = factor(rep(
    c('Full pffr', 'Historical pffr'),
    times = c(nrow(FullModel2), nrow(FullModel2_Hist))
  ))
)
# Separating the dataframes into two
# Unrestricted
Full <- ggplot(data = FullBetasdf[FullBetasdf$Fit == "Full pffr", ]) +
  geom_raster(aes(x = s, y = t, fill = beta)) +
  scale_fill_viridis_c() +
  xlab('s') +
  ylab('t') +
  theme_bw() +
  ggtitle('Unrestricted pffr Surface Fit')
# Historical
HistPlotData <- FullBetasdf[FullBetasdf$Fit == "Historical pffr", ]
# Set beta to NA wherever s >= t
HistPlotData$beta[HistPlotData$s >= HistPlotData$t] <- NA
Full_Hist <- ggplot(data = HistPlotData) +
  geom_raster(aes(x = s, y = t, fill = beta)) +
  scale_fill_viridis_c(na.value = "white") +
  xlab('s') +
  ylab('t') +
  theme_bw() +
  ggtitle('Historical pffr Surface Fit')
grid.arrange(Full, Full_Hist)
area_train <- fires_month |>
  filter(year <= 2015)
area_test <- fires_month |>
  filter(year >= 2016)
fires_train <- fires_counts |>
  filter(year <= 2015)
fires_test <- fires_counts |>

```

```

    filter(year >= 2016)
## Y Matricies
# Make the training dataset a matrix for the model
Y_dat_area_train <- area_train |>
  pivot_wider(
    names_from = month,
    values_from = total_size
  )
Y_area_train <- as.matrix(Y_dat_area_train)[, -1]
# Make the test dataset a matrix
Y_dat_area_test <- area_test |>
  pivot_wider(
    names_from = month,
    values_from = total_size
  )
Y_area_test <- as.matrix(Y_dat_area_test)[, -1]
## X matrices
# Make the training dataset a matrix for the model
X_dat_fires_train <- fires_train |>
  pivot_wider(
    names_from = month,
    values_from = n_fires
  )
X_fires_train <- as.matrix(X_dat_fires_train)[, -1]
# Make the test dataset a matrix
X_dat_fires_test <- fires_test |>
  pivot_wider(
    names_from = month,
    values_from = n_fires
  )
X_fires_test <- as.matrix(X_dat_fires_test)[, -1]
t <- 1:ncol(Y_area_train) # response index (months)
s <- 1:ncol(X_fires_train) # predictor index (months)
# For total area burned take the log
Y_log_train <- log(Y_area_train + 1)
# For fire counts take the log
X_log_train <- log(X_fires_train + 1)
Train_Model <- pffr(
  Y_log_train ~ ff(X_log_train, xind = s),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))

```

```

)
Train_FullBetas <- coef(Train_Model, n1 = 12, n2 = 12)$smterms[[2]]$value
R2_full_train <- summary(Train_Model)
# R^2 = 0.613
Train_Model_Hist <- pffr(
  Y_log_train ~ ff(X_log_train, xind = s, limits = 's<t'),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)
Train_HistBetas <- coef(Train_Model_Hist, n1 = 12, n2 = 12)$smterms[[2]]$value
R2_hist_train <- summary(Train_Model_Hist)
# R-squared of .637
X_log_test <- log(X_fires_test + 1)
reg.train_preds <- predict(
  Train_Model,
  newdata = list(X_log_train = X_log_test)
)
# Look at RMSE
y_test <- t(Y_area_test)
reg.train_preds_t <- t(reg.train_preds)
regular_rmse <- sqrt(mean((y_test - reg.train_preds_t)^2))
regular_rmse
# Get intercept
smterms <- coef(Train_Model_Hist, n1 = 12, n2 = 12)$smterms[[1]]$value
pterm <- coef(Train_Model_Hist, n1 = 12, n2 = 12)$pterm[1, 1]
b0 <- pterm + smterms
b0_matrix <- matrix(b0, nrow = 12, ncol = 5)
# Get the beta surface matrix (12 x 12)
b1 <- Train_HistBetas
b1 <- matrix(b1, nrow = 12, ncol = 12)
b1[lower.tri(b1, diag = TRUE)] <- 0 # taking out values where s>=t
# Manual prediction
b1_matrix <- X_log_test %*% b1
b1_matrix_t <- t(b1_matrix)
hist.train_preds <- b0_matrix + b1_matrix_t
## Get RMSE's
historical_rmse <- sqrt(mean((y_test - hist.train_preds)^2))
historical_rmse
test_years <- 2016:2020
Y_log_test <- log(Y_area_test + 1)
# --- Actual values (long format)

```

```

actual_long <- as.data.frame(Y_log_test) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "actual") |>
  mutate(month = as.numeric(month))
# Then redo the long format with back-transformed values
reg_long <- as.data.frame(reg.train_preds) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "pred_reg") |>
  mutate(month = as.numeric(month))
hist_long <- as.data.frame(t(hist.train_preds)) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "pred_hist") |>
  mutate(month = as.numeric(month))
# Re-join and replot
plot_dat <- actual_long |>
  left_join(reg_long, by = c("year", "month")) |>
  left_join(hist_long, by = c("year", "month")) |>
  mutate(year = factor(year))
p1 <- ggplot(plot_dat, aes(x = month, color = year, group = year)) +
  geom_line(aes(y = actual), linewidth = 0.8) +
  geom_line(aes(y = pred_reg), linetype = "dashed", linewidth = 0.8) +
  geom_line(aes(y = pred_hist), linetype = "dotted", linewidth = 0.8) +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  labs(
    title = "Actual (solid) vs Unrestricted (dashed) vs Historical (dotted) Predictions",
    x = "Month",
    y = "log(Mean Fire Size)",
    color = "Year"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
Train_Model_n1 <- pffr(
  Y_area_train ~ ff(X_fires_train, xind = s),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)

Train_FullBetas_n1 <- coef(Train_Model_n1, n1 = 12, n2 = 12)$smterms[[2]]$value

```



```

R2_full_train_nl <- summary(Train_Model_nl)
# R^2 = 0.292

Train_Model_Hist_nl <- pffr(
  Y_area_train ~ ff(X_fires_train, xind = s, limits = 's<t'),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)

Train_HistBetas_nl <- coef(Train_Model_Hist_nl, n1 = 12, n2 = 12)$smterms[[
  2
]]$value

R2_hist_train_nl <- summary(Train_Model_Hist_nl)
# R-squared of .321

reg.train_preds_nl <- predict(
  Train_Model_nl,
  newdata = list(X_fires_train = X_fires_test)
)

# Look at RMSE
y_test <- t(Y_area_test)
reg.train_preds_t_nl <- t(reg.train_preds_nl)

regular_rmse_nl <- sqrt(mean((y_test - reg.train_preds_t_nl)^2))
regular_rmse_nl
#104.20

# Get intercept
smterms <- coef(Train_Model_Hist_nl, n1 = 12, n2 = 12)$smterms[[1]]$value
pters <- coef(Train_Model_Hist_nl, n1 = 12, n2 = 12)$pters[1, 1]
b0 <- pters + smterms
b0_matrix <- matrix(b0, nrow = 12, ncol = 5)

# Get the beta surface matrix (12 x 12)
b1 <- Train_HistBetas_nl
b1 <- matrix(b1, nrow = 12, ncol = 12)
b1[lower.tri(b1, diag = TRUE)] <- 0 # taking out values where s>=t

# Manual prediction

```

```

b1_matrix <- X_fires_test %*% b1
b1_matrix_t <- t(b1_matrix)

hist.train_preds_nl <- b0_matrix + b1_matrix_t

## Get RMSE's
historical_rmse_nl <- sqrt(mean((y_test - hist.train_preds_nl)^2))
historical_rmse_nl
# 103.58

test_years <- 2016:2020

# --- Actual values (long format)
actual_long <- as.data.frame(Y_area_test) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "actual") |>
  mutate(month = as.numeric(month))

# Then redo the long format with back-transformed values
reg_long <- as.data.frame(reg.train_preds_nl) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "pred_reg") |>
  mutate(month = as.numeric(month))

hist_long <- as.data.frame(t(hist.train_preds_nl)) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "pred_hist") |>
  mutate(month = as.numeric(month))

# Re-join and replot
plot_dat <- actual_long |>
  left_join(reg_long, by = c("year", "month")) |>
  left_join(hist_long, by = c("year", "month")) |>
  mutate(year = factor(year))

p2 <- ggplot(plot_dat, aes(x = month, color = year, group = year)) +
  geom_line(aes(y = actual), linewidth = 0.8) +
  geom_line(aes(y = pred_reg), linetype = "dashed", linewidth = 0.8) +
  geom_line(aes(y = pred_hist), linetype = "dotted", linewidth = 0.8) +

```

```

scale_x_continuous(breaks = 1:12, labels = month.abb) +
labs(
  title = "Actual (solid) vs Unrestricted (dashed) vs Historical (dotted)",
  x = "Month",
  y = "Mean Fire Size",
  color = "Year"
) +
theme_minimal() +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
cowplot::plot_grid(p1, p2, ncol = 1, rel_heights = c(25, 25))
ggplot(
  fires_counts,
  aes(x = month, y = n_fires, color = factor(year), group = year)
) +
geom_line() +
labs(
  title = "Fire counts per month",
  x = "Month",
  y = "Number of fires",
  color = "Year"
) +
theme_minimal()

ggplot(
  fires_month,
  aes(x = month, y = total_size, color = factor(year), group = year)
) +
geom_line() +
labs(
  title = "Total area burned per month",
  x = "Month",
  y = "Total FIRE_SIZE",
  color = "Year"
) +
theme_minimal()
library(ggplot2)
library(tidyverse)
library(refund)
library(gridExtra)
library(patchwork)

fires <- vroom::vroom("forestfires.csv")

```

```

fires_counts <- fires |>
  mutate(
    year = FIRE_YEAR,
    month = month(DISCOVERY_DATE, label = TRUE)
  ) |>
  count(year, month, name = "n_fires")

fires_month <- fires |>
  mutate(
    year = FIRE_YEAR,
    month = month(DISCOVERY_DATE, label = TRUE)
  ) |>
  group_by(year, month) |>
  summarise(total_size = mean(FIRE_SIZE, na.rm = TRUE), .groups = "drop")

ggplot(
  fires_counts,
  aes(x = month, y = n_fires, color = factor(year), group = year)
) +
  geom_line() +
  labs(
    title = "Fire counts per month",
    x = "Month",
    y = "Number of fires",
    color = "Year"
  ) +
  theme_minimal()

ggplot(
  fires_month,
  aes(x = month, y = total_size, color = factor(year), group = year)
) +
  geom_line() +
  labs(
    title = "Total area burned per month",
    x = "Month",
    y = "Total FIRE_SIZE",
    color = "Year"
  ) +
  theme_minimal()

# Making the dataset for our Y matrix

```

```

# Taking the values from month column and makes them each their own column
Y_dat <- fires_month |>
  pivot_wider(
    names_from = month,
    values_from = total_size
  )
# Then I want to make this a matrix without the year variable
Y_mat <- as.matrix(Y_dat)[, -1]
# Then do the same for the X matrix
# Create the dataset
X_dat <- fires_counts |>
  pivot_wider(
    names_from = month,
    values_from = n_fires
  )
# Then I want to make this a matrix
X_mat <- as.matrix(X_dat)[, -1]
t <- 1:ncol(Y_mat) # response index (months)
s <- 1:ncol(X_mat) # predictor index (months)
# For total area burned take the log
Y_log <- log(Y_mat + 1)
# For fire counts take the log
X_log <- log(X_mat + 1)

FullModel <- pffr(
  Y_log ~ ff(X_log, xind = s),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)
FullBetas <- coef(FullModel, n1 = 12, n2 = 12)$smterms[[2]]$value
R2_reg <- summary(FullModel)
# R-squared of .568

FullModel_Hist <- pffr(
  Y_log ~ ff(X_log, xind = s, limits = 's<t'),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)
FullHistBetas <- coef(FullModel_Hist, n1 = 12, n2 = 12)$smterms[[2]]$value
R2_hist <- summary(FullModel_Hist)

```

```

# R-squared of .6

FullModel2 <- coef(FullModel, n1 = 12, n2 = 12)$smterms[[2]]$coef
FullModel2_Hist <- coef(FullModel_Hist, n1 = 12, n2 = 12)$smterms[[2]]$coef
# Creating the dataframe
FullBetasdf <- data.frame(
  't' = c(FullModel2$X_log.tmat, FullModel2_Hist$X_log.tmat),
  's' = c(FullModel2$X_log.smat, FullModel2_Hist$X_log.smat),
  'beta' = c(FullModel2$value, FullModel2_Hist$value),
  'Fit' = factor(rep(
    c('Full pffr', 'Historical pffr'),
    times = c(nrow(FullModel2), nrow(FullModel2_Hist))
  ))
)
# Separating the dataframes into two
# Unrestricted
Full <- ggplot(data = FullBetasdf[FullBetasdf$Fit == "Full pffr", ]) +
  geom_raster(aes(x = s, y = t, fill = beta)) +
  scale_fill_viridis_c() +
  xlab('s') +
  ylab('t') +
  theme_bw() +
  ggtitle('Unrestricted pffr Surface Fit')
# Historical
HistPlotData <- FullBetasdf[FullBetasdf$Fit == "Historical pffr", ]
# Set beta to NA wherever s >= t
HistPlotData$beta[HistPlotData$s >= HistPlotData$t] <- NA
Full_Hist <- ggplot(data = HistPlotData) +
  geom_raster(aes(x = s, y = t, fill = beta)) +
  scale_fill_viridis_c(na.value = "white") +
  xlab('s') +
  ylab('t') +
  theme_bw() +
  ggtitle('Historical pffr Surface Fit')
grid.arrange(Full, Full_Hist)

area_train <- fires_month |>
  filter(year <= 2015)
area_test <- fires_month |>
  filter(year >= 2016)
fires_train <- fires_counts |>
  filter(year <= 2015)

```

```

fires_test <- fires_counts |>
  filter(year >= 2016)
## Y Matricies
# Make the training dataset a matrix for the model
Y_dat_area_train <- area_train |>
  pivot_wider(
    names_from = month,
    values_from = total_size
  )
Y_area_train <- as.matrix(Y_dat_area_train)[, -1]
# Make the test dataset a matrix
Y_dat_area_test <- area_test |>
  pivot_wider(
    names_from = month,
    values_from = total_size
  )
Y_area_test <- as.matrix(Y_dat_area_test)[, -1]
## X matrices
# Make the training dataset a matrix for the model
X_dat_fires_train <- fires_train |>
  pivot_wider(
    names_from = month,
    values_from = n_fires
  )
X_fires_train <- as.matrix(X_dat_fires_train)[, -1]
# Make the test dataset a matrix
X_dat_fires_test <- fires_test |>
  pivot_wider(
    names_from = month,
    values_from = n_fires
  )
X_fires_test <- as.matrix(X_dat_fires_test)[, -1]
t <- 1:ncol(Y_area_train) # response index (months)
s <- 1:ncol(X_fires_train) # predictor index (months)
# For total area burned take the log
Y_log_train <- log(Y_area_train + 1)
# For fire counts take the log
X_log_train <- log(X_fires_train + 1)

Train_Model <- pffr(
  Y_log_train ~ ff(X_log_train, xind = s),
  yind = t,

```

```

method = "NCV",
bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)
Train_FullBetas <- coef(Train_Model, n1 = 12, n2 = 12)$smterms[[2]]$value
R2_full_train <- summary(Train_Model)
# R^2 = 0.613

Train_Model_Hist <- pffr(
  Y_log_train ~ ff(X_log_train, xind = s, limits = 's<t'),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)
Train_HistBetas <- coef(Train_Model_Hist, n1 = 12, n2 = 12)$smterms[[2]]$value
R2_hist_train <- summary(Train_Model_Hist)
# R-squared of .637

X_log_test <- log(X_fires_test + 1)
reg.train_preds <- predict(
  Train_Model,
  newdata = list(X_log_train = X_log_test)
)
# Look at RMSE
y_test <- t(Y_area_test)
reg.train_preds_t <- t(reg.train_preds)
regular_rmse <- sqrt(mean((y_test - reg.train_preds_t)^2))
regular_rmse

## We need to do this manually
## Predict function doesn't work with limits
# Get intercept
smterms <- coef(Train_Model_Hist, n1 = 12, n2 = 12)$smterms[[1]]$value
pterm <- coef(Train_Model_Hist, n1 = 12, n2 = 12)$pterm[1, 1]
b0 <- pterm + smterms
b0_matrix <- matrix(b0, nrow = 12, ncol = 5)
# Get the beta surface matrix (12 x 12)
b1 <- Train_HistBetas
b1 <- matrix(b1, nrow = 12, ncol = 12)
b1[lower.tri(b1, diag = TRUE)] <- 0 # taking out values where s>=t
# Manual prediction
b1_matrix <- X_log_test %*% b1
b1_matrix_t <- t(b1_matrix)

```



```

hist.train_preds <- b0_matrix + b1_matrix_t
## Get RMSE's
historical_rmse <- sqrt(mean((y_test - hist.train_preds)^2))
historical_rmse

test_years <- 2016:2020
Y_log_test <- log(Y_area_test + 1)
# --- Actual values (long format)
actual_long <- as.data.frame(Y_log_test) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "actual") |>
  mutate(month = as.numeric(month))
# Then redo the long format with back-transformed values
reg_long <- as.data.frame(reg.train_preds) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "pred_reg") |>
  mutate(month = as.numeric(month))
hist_long <- as.data.frame(t(hist.train_preds)) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "pred_hist") |>
  mutate(month = as.numeric(month))
# Re-join and replot
plot_dat <- actual_long |>
  left_join(reg_long, by = c("year", "month")) |>
  left_join(hist_long, by = c("year", "month")) |>
  mutate(year = factor(year))
p1 <- ggplot(plot_dat, aes(x = month, color = year, group = year)) +
  geom_line(aes(y = actual), linewidth = 0.8) +
  geom_line(aes(y = pred_reg), linetype = "dashed", linewidth = 0.8) +
  geom_line(aes(y = pred_hist), linetype = "dotted", linewidth = 0.8) +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  labs(
    title = "Actual (solid) vs Unrestricted (dashed) vs Historical (dotted) Predictions",
    x = "Month",
    y = "log(Mean Fire Size)",
    color = "Year"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

```

Train_Model_nl <- pffr(
  Y_area_train ~ ff(X_fires_train, xind = s),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)

Train_FullBetas_nl <- coef(Train_Model_nl, n1 = 12, n2 = 12)$smterms[[2]]$value

R2_full_train_nl <- summary(Train_Model_nl)
# R^2 = 0.292

Train_Model_Hist_nl <- pffr(
  Y_area_train ~ ff(X_fires_train, xind = s, limits = 's<t'),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)

Train_HistBetas_nl <- coef(Train_Model_Hist_nl, n1 = 12, n2 = 12)$smterms[[
  2
]]$value

R2_hist_train_nl <- summary(Train_Model_Hist_nl)
# R-squared of .321

reg.train_preds_nl <- predict(
  Train_Model_nl,
  newdata = list(X_fires_train = X_fires_test)
)

# Look at RMSE
y_test <- t(Y_area_test)
reg.train_preds_t_nl <- t(reg.train_preds_nl)

regular_rmse_nl <- sqrt(mean((y_test - reg.train_preds_t_nl)^2))
regular_rmse_nl
#104.20

# Get intercept
smterms <- coef(Train_Model_Hist_nl, n1 = 12, n2 = 12)$smterms[[1]]$value
pterm <- coef(Train_Model_Hist_nl, n1 = 12, n2 = 12)$pterm[1, 1]

```

```

b0 <- pterms + smterms
b0_matrix <- matrix(b0, nrow = 12, ncol = 5)

# Get the beta surface matrix (12 x 12)
b1 <- Train_HistBetas_nl
b1 <- matrix(b1, nrow = 12, ncol = 12)
b1[lower.tri(b1, diag = TRUE)] <- 0 # taking out values where s>=t

# Manual prediction
b1_matrix <- X_fires_test %*% b1
b1_matrix_t <- t(b1_matrix)

hist.train_preds_nl <- b0_matrix + b1_matrix_t

## Get RMSE's
historical_rmse_nl <- sqrt(mean((y_test - hist.train_preds_nl)^2))
historical_rmse_nl
# 103.58

test_years <- 2016:2020

# --- Actual values (long format)
actual_long <- as.data.frame(Y_area_test) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "actual") |>
  mutate(month = as.numeric(month))

# Then redo the long format with back-transformed values
reg_long <- as.data.frame(reg.train_preds_nl) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "pred_reg") |>
  mutate(month = as.numeric(month))

hist_long <- as.data.frame(t(hist.train_preds_nl)) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "pred_hist") |>
  mutate(month = as.numeric(month))

# Re-join and replot

```

```

plot_dat <- actual_long |>
  left_join(reg_long, by = c("year", "month")) |>
  left_join(hist_long, by = c("year", "month")) |>
  mutate(year = factor(year))

p2 <- ggplot(plot_dat, aes(x = month, color = year, group = year)) +
  geom_line(aes(y = actual), linewidth = 0.8) +
  geom_line(aes(y = pred_reg), linetype = "dashed", linewidth = 0.8) +
  geom_line(aes(y = pred_hist), linetype = "dotted", linewidth = 0.8) +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  labs(
    title = "Actual (solid) vs Unrestricted (dashed) vs Historical (dotted)",
    x = "Month",
    y = "Mean Fire Size",
    color = "Year"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
cowplot::plot_grid(p1, p2, ncol = 1, rel_heights = c(25, 25))

```

```

library(ggplot2)
library(tidyverse)
library(refund)
library(gridExtra)
library(patchwork)

fires <- vroom::vroom("forestfires.csv")

fires_counts <- fires |>
  mutate(
    year = FIRE_YEAR,
    month = month(DISCOVERY_DATE, label = TRUE)
  ) |>
  count(year, month, name = "n_fires")

fires_month <- fires |>
  mutate(
    year = FIRE_YEAR,
    month = month(DISCOVERY_DATE, label = TRUE)
  ) |>
  group_by(year, month) |>
  summarise(total_size = mean(FIRE_SIZE, na.rm = TRUE), .groups = "drop")

```

```

ggplot(
  fires_counts,
  aes(x = month, y = n_fires, color = factor(year), group = year)
) +
  geom_line() +
  labs(
    title = "Fire counts per month",
    x = "Month",
    y = "Number of fires",
    color = "Year"
  ) +
  theme_minimal()

```

```

ggplot(
  fires_month,
  aes(x = month, y = total_size, color = factor(year), group = year)
) +
  geom_line() +
  labs(
    title = "Total area burned per month",
    x = "Month",
    y = "Total FIRE_SIZE",
    color = "Year"
  ) +
  theme_minimal()

```

```

# Making the dataset for our Y matrix
# Taking the values from month column and makes them each their own column
Y_dat <- fires_month |>
  pivot_wider(
    names_from = month,
    values_from = total_size
  )
# Then I want to make this a matrix without the year variable
Y_mat <- as.matrix(Y_dat)[, -1]
# Then do the same for the X matrix
# Create the dataset
X_dat <- fires_counts |>
  pivot_wider(
    names_from = month,
    values_from = n_fires
  )

```

```

)
# Then I want to make this a matrix
X_mat <- as.matrix(X_dat)[, -1]
t <- 1:ncol(Y_mat) # response index (months)
s <- 1:ncol(X_mat) # predictor index (months)
# For total area burned take the log
Y_log <- log(Y_mat + 1)
# For fire counts take the log
X_log <- log(X_mat + 1)

FullModel <- pffr(
  Y_log ~ ff(X_log, xind = s),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)
FullBetas <- coef(FullModel, n1 = 12, n2 = 12)$smterms[[2]]$value
R2_reg <- summary(FullModel)
# R-squared of .568

FullModel_Hist <- pffr(
  Y_log ~ ff(X_log, xind = s, limits = 's<t'),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)
FullHistBetas <- coef(FullModel_Hist, n1 = 12, n2 = 12)$smterms[[2]]$value
R2_hist <- summary(FullModel_Hist)
# R-squared of .6

FullModel2 <- coef(FullModel, n1 = 12, n2 = 12)$smterms[[2]]$coef
FullModel2_Hist <- coef(FullModel_Hist, n1 = 12, n2 = 12)$smterms[[2]]$coef
# Creating the dataframe
FullBetasdf <- data.frame(
  't' = c(FullModel2$X_log.tmat, FullModel2_Hist$X_log.tmat),
  's' = c(FullModel2$X_log.smat, FullModel2_Hist$X_log.smat),
  'beta' = c(FullModel2$value, FullModel2_Hist$value),
  'Fit' = factor(rep(
    c('Full pffr', 'Historical pffr'),
    times = c(nrow(FullModel2), nrow(FullModel2_Hist))
  ))
)

```

```

# Separating the dataframes into two
# Unrestricted
Full <- ggplot(data = FullBetasdf[FullBetasdf$Fit == "Full pffr", ]) +
  geom_raster(aes(x = s, y = t, fill = beta)) +
  scale_fill_viridis_c() +
  xlab('s') +
  ylab('t') +
  theme_bw() +
  ggtitle('Unrestricted pffr Surface Fit')
# Historical
HistPlotData <- FullBetasdf[FullBetasdf$Fit == "Historical pffr", ]
# Set beta to NA wherever s >= t
HistPlotData$beta[HistPlotData$s >= HistPlotData$t] <- NA
Full_Hist <- ggplot(data = HistPlotData) +
  geom_raster(aes(x = s, y = t, fill = beta)) +
  scale_fill_viridis_c(na.value = "white") +
  xlab('s') +
  ylab('t') +
  theme_bw() +
  ggtitle('Historical pffr Surface Fit')
grid.arrange(Full, Full_Hist)

```

```

area_train <- fires_month |>
  filter(year <= 2015)
area_test <- fires_month |>
  filter(year >= 2016)
fires_train <- fires_counts |>
  filter(year <= 2015)
fires_test <- fires_counts |>
  filter(year >= 2016)
## Y Matrices
# Make the training dataset a matrix for the model
Y_dat_area_train <- area_train |>
  pivot_wider(
    names_from = month,
    values_from = total_size
  )
Y_area_train <- as.matrix(Y_dat_area_train)[, -1]
# Make the test dataset a matrix
Y_dat_area_test <- area_test |>
  pivot_wider(
    names_from = month,

```

```

    values_from = total_size
  )
Y_area_test <- as.matrix(Y_dat_area_test)[, -1]
## X matrices
# Make the training dataset a matrix for the model
X_dat_fires_train <- fires_train |>
  pivot_wider(
    names_from = month,
    values_from = n_fires
  )
X_fires_train <- as.matrix(X_dat_fires_train)[, -1]
# Make the test dataset a matrix
X_dat_fires_test <- fires_test |>
  pivot_wider(
    names_from = month,
    values_from = n_fires
  )
X_fires_test <- as.matrix(X_dat_fires_test)[, -1]
t <- 1:ncol(Y_area_train) # response index (months)
s <- 1:ncol(X_fires_train) # predictor index (months)
# For total area burned take the log
Y_log_train <- log(Y_area_train + 1)
# For fire counts take the log
X_log_train <- log(X_fires_train + 1)

Train_Model <- pffr(
  Y_log_train ~ ff(X_log_train, xind = s),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)
Train_FullBetas <- coef(Train_Model, n1 = 12, n2 = 12)$smterms[[2]]$value
R2_full_train <- summary(Train_Model)
# R^2 = 0.613

Train_Model_Hist <- pffr(
  Y_log_train ~ ff(X_log_train, xind = s, limits = 's<t'),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)
Train_HistBetas <- coef(Train_Model_Hist, n1 = 12, n2 = 12)$smterms[[2]]$value

```



```

R2_hist_train <- summary(Train_Model_Hist)
# R-squared of .637

X_log_test <- log(X_fires_test + 1)
reg.train_preds <- predict(
  Train_Model,
  newdata = list(X_log_train = X_log_test)
)
# Look at RMSE
y_test <- t(Y_area_test)
reg.train_preds_t <- t(reg.train_preds)
regular_rmse <- sqrt(mean((y_test - reg.train_preds_t)^2))
regular_rmse

## We need to do this manually
## Predict function doesn't work with limits
# Get intercept
smterms <- coef(Train_Model_Hist, n1 = 12, n2 = 12)$smterms[[1]]$value
ptersms <- coef(Train_Model_Hist, n1 = 12, n2 = 12)$ptersms[1, 1]
b0 <- ptersms + smterms
b0_matrix <- matrix(b0, nrow = 12, ncol = 5)
# Get the beta surface matrix (12 x 12)
b1 <- Train_HistBetas
b1 <- matrix(b1, nrow = 12, ncol = 12)
b1[lower.tri(b1, diag = TRUE)] <- 0 # taking out values where s>=t
# Manual prediction
b1_matrix <- X_log_test %*% b1
b1_matrix_t <- t(b1_matrix)
hist.train_preds <- b0_matrix + b1_matrix_t
## Get RMSE's
historical_rmse <- sqrt(mean((y_test - hist.train_preds)^2))
historical_rmse

test_years <- 2016:2020
Y_log_test <- log(Y_area_test + 1)
# --- Actual values (long format)
actual_long <- as.data.frame(Y_log_test) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "actual") |>
  mutate(month = as.numeric(month))
# Then redo the long format with back-transformed values

```

```

reg_long <- as.data.frame(reg.train_preds) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "pred_reg") |>
  mutate(month = as.numeric(month))
hist_long <- as.data.frame(t(hist.train_preds)) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "pred_hist") |>
  mutate(month = as.numeric(month))
# Re-join and replot
plot_dat <- actual_long |>
  left_join(reg_long, by = c("year", "month")) |>
  left_join(hist_long, by = c("year", "month")) |>
  mutate(year = factor(year))
p1 <- ggplot(plot_dat, aes(x = month, color = year, group = year)) +
  geom_line(aes(y = actual), linewidth = 0.8) +
  geom_line(aes(y = pred_reg), linetype = "dashed", linewidth = 0.8) +
  geom_line(aes(y = pred_hist), linetype = "dotted", linewidth = 0.8) +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  labs(
    title = "Actual (solid) vs Unrestricted (dashed) vs Historical (dotted) Predictions",
    x = "Month",
    y = "log(Mean Fire Size)",
    color = "Year"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

Train_Model_n1 <- pffr(
  Y_area_train ~ ff(X_fires_train, xind = s),
  yind = t,
  method = "NCV",
  bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)

Train_FullBetas_n1 <- coef(Train_Model_n1, n1 = 12, n2 = 12)$smterms[[2]]$value

R2_full_train_n1 <- summary(Train_Model_n1)
# R^2 = 0.292

Train_Model_Hist_n1 <- pffr(

```

```

Y_area_train ~ ff(X_fires_train, xind = s, limits = 's<t'),
yind = t,
method = "NCV",
bs.int = list(bs = "ps", k = 5, m = c(2, 1))
)

Train_HistBetas_nl <- coef(Train_Model_Hist_nl, n1 = 12, n2 = 12)$smterms[[
  2
]]$value

R2_hist_train_nl <- summary(Train_Model_Hist_nl)
# R-squared of .321

reg.train_preds_nl <- predict(
  Train_Model_nl,
  newdata = list(X_fires_train = X_fires_test)
)

# Look at RMSE
y_test <- t(Y_area_test)
reg.train_preds_t_nl <- t(reg.train_preds_nl)

regular_rmse_nl <- sqrt(mean((y_test - reg.train_preds_t_nl)^2))
regular_rmse_nl
#104.20

# Get intercept
smterms <- coef(Train_Model_Hist_nl, n1 = 12, n2 = 12)$smterms[[1]]$value
pterm <- coef(Train_Model_Hist_nl, n1 = 12, n2 = 12)$pterm[1, 1]
b0 <- pterm + smterms
b0_matrix <- matrix(b0, nrow = 12, ncol = 5)

# Get the beta surface matrix (12 x 12)
b1 <- Train_HistBetas_nl
b1 <- matrix(b1, nrow = 12, ncol = 12)
b1[lower.tri(b1, diag = TRUE)] <- 0 # taking out values where s>=t

# Manual prediction
b1_matrix <- X_fires_test %*% b1
b1_matrix_t <- t(b1_matrix)

hist.train_preds_nl <- b0_matrix + b1_matrix_t

```

```

## Get RMSE's
historical_rmse_nl <- sqrt(mean((y_test - hist.train_preds_nl)^2))
historical_rmse_nl
# 103.58

test_years <- 2016:2020

# --- Actual values (long format)
actual_long <- as.data.frame(Y_area_test) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "actual") |>
  mutate(month = as.numeric(month))

# Then redo the long format with back-transformed values
reg_long <- as.data.frame(reg.train_preds_nl) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "pred_reg") |>
  mutate(month = as.numeric(month))

hist_long <- as.data.frame(t(hist.train_preds_nl)) |>
  setNames(1:12) |>
  mutate(year = test_years) |>
  pivot_longer(-year, names_to = "month", values_to = "pred_hist") |>
  mutate(month = as.numeric(month))

# Re-join and replot
plot_dat <- actual_long |>
  left_join(reg_long, by = c("year", "month")) |>
  left_join(hist_long, by = c("year", "month")) |>
  mutate(year = factor(year))

p2 <- ggplot(plot_dat, aes(x = month, color = year, group = year)) +
  geom_line(aes(y = actual), linewidth = 0.8) +
  geom_line(aes(y = pred_reg), linetype = "dashed", linewidth = 0.8) +
  geom_line(aes(y = pred_hist), linetype = "dotted", linewidth = 0.8) +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  labs(
    title = "Actual (solid) vs Unrestricted (dashed) vs Historical (dotted)",
    x = "Month",
    y = "Mean Fire Size",

```

```

    color = "Year"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
cowplot::plot_grid(p1, p2, ncol = 1, rel_heights = c(25, 25))

```

Project Proposal

Our dataset consists of wildland fire incidents in the United States, with each row representing a single fire event. The functional variables of interest are those that vary over time or along the fire's progression: fire size (FIRE_SIZE), the discovery date (DISCOVERY_DATE, which we aggregate to months), spatial coordinates (X latitude and Y longitude), and the containment date (CONT_DATE) that marks when the fire was brought under control. These variables allow us to construct monthly trajectories of burned area and to view fire activity as a spatio-temporal process rather than a static collection of events. The non-functional variables provide contextual information that helps explain these trajectories. Key fields include administrative location (STATE, COUNTY), cause descriptors (NWCG_CAUSE_CLASSIFICATION, NWCG_GENERAL_CAUSE), temporal grouping (FIRE_YEAR), and categorical fire size (FIRE_SIZE_CLASS). Together, these variables let us examine how patterns in fire size and timing depend on where fires occur, how they start, and whether they fall into smaller (class A–C) versus larger (class D–G) size classes, setting up more detailed functional analyses of regional and cause-specific fire regimes. Our exploratory plots highlight several important patterns. Monthly mean fire size by year shows clear seasonality: in most years average burned area is low in winter and peaks during the late spring–summer fire season, with notable between-year differences in both the height and timing of the seasonal peak.

The companion plot of monthly fire counts reveals that the number of fires also has a strong annual cycle, but the months with the most fires are not always the months with the largest average fire size, suggesting that frequency and severity may be driven by partially different processes (for example, many small human-caused fires versus fewer large landscape-scale events). The spatial map of ignition locations, colored from white (class A, very small fires) to dark red (class G, very large fires), shows fires concentrated in known fire-prone regions such as the Western U.S. mountains and parts of the Southeast, and suggests geographic clusters where large fires are relatively more common.

Finally, the bar plot of NWCG_GENERAL_CAUSE indicates that a few ignition categories dominate overall fire counts, with human-related causes clearly visible alongside lightning and other natural sources, motivating questions about how cause distributions vary across space, time, and fire-size classes. 1. How has the seasonal trajectory of total burned area changed across years, and are there detectable long-term trends in fire severity? Monthly total burned area, aggregated from individual fire size records, constitutes the primary functional response of interest. The covariate of interest here is FIRE_YEAR, which allows each year's twelve-month

burned-area profile to be treated as a single functional observation and compared across time. We can then ask whether the shape of the seasonal curve (its peak timing, amplitude, and duration) has shifted systematically over the decades covered by the dataset. This question is scientifically important because changes in the fire season's intensity and length are among the most direct ecological signatures of climate change. A fire manager or land agency scientist would want to know not only whether fires are getting larger on average, but whether the window of high risk is expanding, since that information directly informs staffing, resource pre-positioning, and prescribed burn scheduling.

2. Does the monthly frequency of fire ignitions follow the same seasonal pattern as burned area, or do fire count and fire severity exhibit distinct temporal dynamics? While Question 1 focuses on area burned, this question treats the monthly count of fire ignitions as a separate functional variable, again with FIRE_YEAR as the primary covariate so that annual ignition-frequency curves can be compared to annual burned-area curves. The exploratory plots already hint that frequency and severity are not perfectly synchronized: some months accumulate many small fires while others see fewer but far larger events. Formally characterizing this divergence and testing whether it varies by year would help distinguish between regimes dominated by high-frequency, low-severity ignitions (often associated with human activity) and those characterized by infrequent but catastrophic fires (more commonly driven by climate and fuel accumulation). For the scientist who collected this data, resolving this distinction has direct implications for cause-attribution modeling and risk forecasting.

3. Does the seasonal profile of burned area differ across geographic regions or states, and can spatially distinct fire regimes be identified from the functional data? This question introduces STATE (or a broader regional grouping) as the key non-functional covariate, and asks whether the monthly burned-area trajectories differ meaningfully across space. The spatial map of ignition locations already suggests geographic clustering of large fires in the Western U.S. and parts of the Southeast, but a functional analysis would go further by comparing the full seasonal curve, not just summary statistics, across regions. Identifying spatially distinct fire regimes matters for resource allocation: a federal land management agency needs to know whether the peak fire season in the Southwest precedes or overlaps with that of the Pacific Northwest, since simultaneous demand for suppression resources across regions creates logistical bottlenecks. State-level functional comparisons could also reveal whether certain jurisdictions have experienced disproportionate increases in large-fire activity over time.

4. Does the cause of ignition, whether natural or human-related, systematically alter the distribution or seasonal trajectory of fire size, and has this relationship changed over time? The covariates in this question are NWCG_GENERAL_CAUSE, as a categorical grouping variable, and FIRE_YEAR, which allows us to assess whether any cause-related differences in fire size or seasonality have strengthened or weakened over time. Together these covariates let us ask whether fires of different origins differ not just in their mean size but in their entire distributional or temporal profile. Human-caused fires may be more uniformly distributed across the calendar year, while lightning-ignited fires may show a sharper seasonal peak tied to monsoonal storm patterns. This question is of direct interest to fire prevention programs, which must decide how to allocate public education and enforcement resources relative to suppression investments, and becomes especially pressing as wildland-urban interface expansion increases human ignition pressure in fire-prone landscapes.

5. How does time-to-containment vary as a

function of fire size class, geographic location, and year, and is there evidence of systematic improvement in suppression efficiency over time? The covariates here are `FIRE_SIZE_CLASS` and `STATE` as non-functional grouping variables, and `FIRE_YEAR` as a temporal covariate, allowing us to examine whether containment duration, computed as the difference between containment day-of-year and discovery day-of-year, differs across size categories, varies by state, and has trended in any consistent direction over time. For the scientist or agency administrator who assembled this dataset, this question has direct operational relevance: if containment times for large fires (class E through G) have not improved despite advances in aerial suppression and early detection technology, that is an important finding that would motivate deeper investigation into resource adequacy and climate-driven fire behavior. Finally, the bar plot of `NWCG_GENERAL_CAUSE` indicates that a few ignition categories dominate overall fire counts, with human-related causes clearly visible alongside lightning and other natural sources, motivating questions about how cause distributions vary across space, time, and fire-size classes.

1. How has the seasonal trajectory of total burned area changed across years, and are there detectable long-term trends in fire severity? Monthly total burned area, aggregated from individual fire size records, constitutes the primary functional response of interest. The covariate of interest here is `FIRE_YEAR`, which allows each year's twelve-month burned-area profile to be treated as a single functional observation and compared across time. We can then ask whether the shape of the seasonal curve (its peak timing, amplitude, and duration) has shifted systematically over the decades covered by the dataset. This question is scientifically important because changes in the fire season's intensity and length are among the most direct ecological signatures of climate change. A fire manager or land agency scientist would want to know not only whether fires are getting larger on average, but whether the window of high risk is expanding, since that information directly informs staffing, resource pre-positioning, and prescribed burn scheduling.
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The covariates here are FIRE_SIZE_CLASS and STATE as non-functional grouping variables, and FIRE_YEAR as a temporal covariate, allowing us to examine whether containment duration, computed as the difference between containment day-of-year and discovery day-of-year, differs across size categories, varies by state, and has trended in any consistent direction over time. For the scientist or agency administrator who assembled this dataset, this question has direct operational relevance: if containment times for large fires (class E through G) have not improved despite advances in aerial suppression and early detection technology, that is an important finding that would motivate deeper investigation into resource adequacy and climate-driven fire behavior.